



Development and validation of an LC-Orbitrap-HRMS method for quantification of human insulin and selected insulin analogues with application on two forensic cases

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ABSTRACT

Insulin, a peptide hormone essential for blood glucose regulation, is typically quantified in clinical practice using immunoassays. While sensitive, these assays lack specificity and are often inadequate for the precise qualification and quantification in forensic practices. To address this limitation, we developed and validated an LC-Orbitrap-HRMS method for the quantification of intact insulin and its analogues in four postmortem matrices: plasma, liver, kidney, and adipose tissue. Method validation demonstrated linearity for human insulin in plasma from 0.1 to 100 ng/mL, with accuracy between 90.1 % and 106.9 %. Notably, in challenging matrices such as adipose tissue, the LLOQ was 1 ng/mL, with accuracy range from 94.1 % to 100.2 %, confirming robust analytical performance. The validated method was applied on two forensic cases involving suspected insulin overdose. Insulin concentrations in subcutaneous adipose tissue from the injection sites were significantly elevated, measuring 743.1 ng/mL and 53.2 ng/mL, respectively. These results demonstrate the utility of the method in providing reliable evidence for the investigation of suspected fatal insulin intoxication.

1. Introduction

Human insulin, the sole endogenous hypoglycemic hormone secreted by pancreatic β -cells, comprises a 21-amino acid A-chain and a 30-amino acid B-chain linked by three disulfide bonds (A7-B7, A20-B19, and A6-A11) [1]. Synthetic insulin analogues, including rapid-acting analogues (e.g., insulin lispro, aspart, and glulisine), and long-acting analogues (e.g., insulin detemir, glargine, and degludec), are developed through structural modifications of the endogenous hormone [2]. These insulin analogues are therapeutically essential, primarily for the treatment of type I diabetes mellitus but also for advanced type II diabetes [3]. However, their potent hypoglycemic effects also increases the risk of severe hypoglycemia and fatal outcomes following intentional misuse or accidental overdose [4]. Increasing global reports of deaths

due to insulin overdoses underscore the need for sensitive and specific analytical methods to quantify insulin analogues in biological samples [5].

In clinical practice, insulin is commonly measured using immunoassays, such as enzyme linked immunosorbent assay (ELISA), radioimmunoassay (RIA), or chemiluminescence immunoassay (CLIA), which provide high sensitivity for total insulin detection in vivo [6]. However, their forensic utility is limited by cross-reactivity and an inability to distinguish structurally similar analogues, compromising specificity where precise qualification and quantification are critical [7]. Liquid chromatography-high-resolution mass spectrometry (LC-HRMS) has thus emerged as the preferred platform for insulin analogues detection in complex biological matrices, due to its superior selectivity, sensitivity, and capacity for multi-analyte analysis [8]. Nevertheless, most existing

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LC-HRMS methods focus on plasma, with scarce data on tissue distribution (e.g., in liver, kidney, or adipose tissue) [9]. In addition, inconsistencies in sample preparation, chromatographic conditions, and mass spectrometry parameters hinder method standardization.

To bridge these gaps, we developed and validated an LC-Orbitrap-HRMS method for the quantification of intact human insulin and synthetic analogues in four postmortem matrices: plasma, liver, kidney, and adipose tissue. The method was successfully applied on two fatal cases of insulin intoxications, demonstrating its potential to provide reliable technical and scientific support for forensic investigations involving insulin overdose.

2. Materials and methods

2.1. Reagents and chemicals

Insulin analogues were selected as representative compounds from major therapeutic categories: aspart insulin (rapid-acting), human insulin (medium-acting), and glargine insulin (long-acting). Porcine insulin served as the internal standard (IS) due to its structural similarity to human insulin and absence in human samples. All insulin standards were obtained as high-purity chemical reference materials from the National Institutes for Food and Drug Control (Beijing, China).

HPLC-grade methanol, isopropanol, acetonitrile, formic acid, and acetic acid were purchased from Thermo Fisher Scientific (Waltham, MA, USA). Concentrated ammonia solution was acquired from Aladdin (Shanghai, China). Solid-phase extraction (SPE) was performed on Oasis® MAX LP Extraction Cartridges (3 cc, 60 mg, Waters, Milford, MA, USA). RIPA lysis buffer and 3-mm stainless steel grinding beads were purchased from Servicebio (Wuhan, China). Nylon membrane filters (0.22 µm) were obtained from Jinteng (Tianjin, China).

Blank postmortem matrices (plasma, liver, kidney, and adipose tissue) were collected from Sprague-Dawley rats that had not been exposed to insulin after anesthesia and euthanasia. This animal study was approved by the Animal Ethics Committee of Huazhong University of Science and Technology (Approval No. 2024-4674). Samples were stored at -80°C and served as blank matrices for method development and validation.

2.2. Preparation of stock and working solutions

Stock solutions (1 mg/mL) were prepared by dissolving each insulin standard in 0.01 mM hydrochloric acid. Working solutions were obtained through 10-fold serial dilution using methanol/water/acetic acid (6:3:1, v/v/v). All solutions were aliquoted (100 µL per vial) and stored at -80°C.

For method development, working solutions were directly infused to optimize instrument parameters. For validation, they were spiked into blank matrices to prepare calibration standards and quality control (QC) samples at desired concentrations. Fresh aliquots were used daily to ensure stability, with no significant degradation observed under these storage conditions.

2.3. Sample preparation

Sample preparation was performed using a modified protocol combining protein precipitation (PPT) and mixed-mode anion-exchange solid-phase extraction (MAX-SPE) as follows [10,11]:

Tissue Homogenization: Tissue samples (200 mg of liver, kidney, or adipose tissue) were homogenized with 1 mL RIPA lysis buffer using two 3-mm stainless steel grinding beads (60 Hz, 2 cycles of 1 min each) in a homogenizer (Jingxin, Shanghai, China). Homogenates were then centrifuged (12,000 rpm, 10 min, 4°C) using a centrifuge (Eppendorf, Hamburg, Germany), and the supernatant was collected.

PPT: A 1 mL volume of tissue homogenate or plasma was mixed with 1 mL ice-cold 80 % methanol containing 200 ng/mL of IS. After

vertexing for 2 min, the mixture was centrifuged (12,000 rpm, 10 min, 4°C). The resulting supernatant was diluted with 2 mL of 5 % NH₄OH.

MAX-SPE: Oasis® MAX LP Extraction Cartridges were preconditioned with 2 mL of methanol followed by 2 mL of ultrapure water. The diluted supernatant was loaded, then washed sequentially with 2 mL 5 % NH₄OH and 2 mL of a solution containing 5 % methanol and 1 % acetic acid. Analytes were eluted with two 500 µL of methanol/water/acetic acid (6:3:1, v/v/v). The combined eluate was concentrated in a vacuum concentrator (Eppendorf, Hamburg, Germany), reconstituted in 200 µL of water containing 0.1 % formic acid, and filtered through a 0.22-µm nylon membrane prior to LC-Orbitrap-HRMS analysis. All solvents were pre-chilled to 4°C before use.

2.4. LC-Orbitrap-HRMS analysis

Chromatographic separation was performed on a Vanquish ultra-high-performance liquid chromatography system (Thermo Fisher Scientific, MA, USA) equipped with an ACQUITY® UPLC Peptide CSH C18 column (100 mm × 2.1 mm, 1.7 µm, Waters, MA, USA). Mobile phases consisted of (A) 0.1 % formic acid in water and (B) 0.1 % formic acid in a mixture of isopropanol/acetonitrile (1:1, v/v). The gradient program was: 5 % B (0 – 1.5 min), 5–30 % B (1.5 – 6 min), 30–95 % B (6 – 8 min), 95 % B (8 – 9.5 min), followed by a return to 5 % B at 9.6 min and re-equilibration until 12 min. Flow rate was 0.3 mL/min, column temperature was maintained at 45°C and injection volume was 20 µL.

Mass spectrometric detection was conducted on an Orbitrap Exploris 240 instrument (Thermo Fisher Scientific, MA, USA) with heated electrospray ionization in positive ion mode (HESI⁺). Source parameters were: spray voltage 3.8 kV, ion transfer tube temperature 320°C, vaporizer temperature 350°C, sheath gas 35 arb, and aux gas 10 arb. Full scan data were acquired at a resolution of 120,000 over the *m/z* range of 400 – 2000 for precursor ion identification, with AGC target set to 1e6 and maximum injection time (IT) set to 100 ms. Quantification was performed in selected-ion monitoring (SIM) mode targeting the most abundant precursor ions at a resolution of 120,000.

3. Method validation

Method validation was conducted in accordance with European Medicines Agency (EMA) and the United States Food and Drug Administration (FDA) guidelines [12,13], assessing specificity, carry-over, matrix effect, extraction recovery, linearity, accuracy and precision. Data were processed using Xcalibur and Origin 2024 software, with statistical analysis performed on GraphPad Prism 8 Software.

3.1. Specificity

Specificity was evaluated by analyzing blank matrices (plasma, liver, kidney, and adipose tissue) obtained from six different individual sources. Blank samples and blanks spiked with IS were processed as described in 2.3. Chromatograms showed no interfering peak at the retention times of target analytes, confirming method specificity.

3.2. Carry-over

Carry-over was assessed by injecting a neat solvent blank (50 % methanol) immediately following the analysis of QC samples (10, 50, and 100 ng/mL). Acceptance required the analyte response in the blank to be less than 10 % of the preceding QC response.

3.3. Matrix effect and extraction recovery

The internal standard-normalized matrix factor (IS-NMF) was calculated using Eq. (1) to assess ionization suppression or enhancement. IS-NMF values within 85 % – 115 % were considered acceptable. Extraction recovery (ER) was determined using Eq. (2). Both parameters

were assessed using three sets of samples prepared in six replicates each from different matrix sources: Group A: Pure standard; Group B: Blank matrix extracts spiked with analyte and IS post-extraction; Group C: Blank matrices spiked with analyte and IS pre-extraction.

$$IS - NMF = \frac{B/A}{B_{IS}/A_{IS}} \quad (1)$$

$$ER(\%) = \frac{C}{B} \times 100 \quad (2)$$

where A, B and C represented the peak areas of analyte in Group A, B, and C, respectively; A_{IS} and B_{IS} denoted the peak areas of IS in Group A and B.

3.4. Linearity

Linearity was evaluated using a nine-point calibration curve (0.1, 0.5, 1, 2, 5, 10, 20, 50, and 100 ng/mL, $n = 6$ per level) by plotting the analyte-to-IS peak area ratio against nominal concentration. As insulin is an endogenous hormone, rather than for mere detection at trace levels, establishing a sufficiently low LLOQ was therefore prioritized to ensure accurate and forensically defensible quantification. Therefore, the lower limit of quantification (LLOQ) was defined as the lowest concentration meeting accuracy (85 % – 115 %) and precision ($RSD \leq 15$ %) criteria. QC samples at LLOQ, low, medium and high concentrations (0.1, 1, 10, and 100, $n = 6$ per level) were analyzed with each calibrations batch to monitor assay performance.

3.5. Accuracy and precision

Accuracy and precision were assessed by analyzing QC samples at four concentrations (0.1, 1, 10, 100 ng/mL; $n = 6$ per level) over three consecutive days. Intra-day performance was evaluated with six replicates per level within one single day; inter-day performance was determined from eighteen replicates per level across all three days. Accuracy was calculated as (measured concentration / nominal concentration) $\times 100$ %, and precision as relative standard deviation (RSD %). Acceptance criteria were ± 15 % for accuracy and ≤ 15 % for RSD %.

4. Forensic application

The validated method was applied to two independent forensic investigations of suspected fatal insulin overdose. Insulin concentrations in subcutaneous adipose tissue (from the suspected injection site), liver, and plasma were quantified using both CLIA and the LC-Orbitrap-HRMS method. All specimens were stored at -80°C before analysis. This study was conducted with informed consent from the deceased's families and approved by the Medical Ethics Committee of Tongji Medical College, Huazhong University of Science and Technology (Approval No. 2025-S102).

5. Results and discussion

5.1. Method development

To establish a sensitive platform for intact insulin and insulin analogues, we compared the performance of three LC-MS systems (triple quadrupole, time-of-flight, and Orbitrap) using insulin standards. As detailed in [Supplementary Materials Section S1](#), the Orbitrap system exhibited superior sensitivity, yielding a signal intensity of 1.27×10^7 for a 100 ng/mL porcine insulin standard, comparable to signals from 100-fold higher concentration on other platforms. This aligns with reported advantages of Orbitrap for analyzing large biomolecules like insulin [14].

HRMS analysis showed distinct charge state distributions among insulin analogues ([Fig. 1](#)). Insulin aspart, human, and porcine predominantly formed $[M+5H]^{5+}$ ions, while glargine insulin showed optimal response as $[M+6H]^{6+}$, consistent with prior trends [15]. Although discrepancies in the dominant charge states have been noted in other studies (e.g., $[M+7H]^{7+}$ for glargine or $[M+6H]^{6+}$ states for Aspart and Human insulin [16,17]), these variations are likely attributable to differences in instrumental parameters rather than intrinsic molecular properties.

Furthermore, all analogues exhibited low dissociation efficiencies (<10 %), attributable to insulin's stable tertiary structure stabilized by three disulfide bonds and a hydrophobic core [18]. This structural stability limits the sensitivity of product ion-based detection methods such as MRM. To overcome this limitation, we implemented the above HRAM strategy targeting intact precursor ions for quantification. This approach eliminates the dependency on fragmentation while maintaining specificity through exact mass filtering. Consequently, the most abundant precursor ion for each analogue (summarized in [Table 1](#)) was selected for quantification in SIM mode.

Current sample preparation approaches for insulin extraction include PPT, SPE, and immunoaffinity purification (IAP), each involving distinct trade-offs in simplicity, selectivity, and cost [19]. PPT is simple and cost-effective but suffers from significant matrix interference, particularly in lipid-rich tissues like adipose [20]. SPE methodologies, particularly mixed-mode cation- or anion-exchange (MCX or MAX) cartridges, utilize pH-dependent charge properties of insulin to enhance selectivity, typically achieving limits of detection near 1 ng/mL [21]. Although high-throughput μ SPE formats are available, their limited sample capacity increases the risk of false negatives in degraded post-mortem samples [15]. IAP can provide exceptional sensitivity ($LLOQ \leq 0.1$ ng/mL) through antibody-based enrichment but remains impractical for routine forensic use due to high cost and procedural complexity [22].

To systematically evaluate these approaches, we compared three preparation strategies using plasma and liver matrices spiked with insulin analogues at 10 ng/mL as samples: PPT alone, PPT with MCX-SPE (PPT+MCX-SPE), and PPT with MAX-SPE (PPT+MAX-SPE). As shown in [Fig. 2](#), PPT+MAX-SPE provided optimal performance, effectively removing matrix interferences while detecting all target analogues. In contrast, glargine insulin was undetectable following PPT+MCX-SPE, likely due to irreversible adsorption. While absolute recoveries were moderate, this limitation was offset by increasing the sample intake to 200 mg of tissue. Critically, the PPT+MAX-SPE method demonstrated consistent performance across all four matrices, including adipose tissue, where insulin accumulates to forensically significant levels at injection sites.

5.2. Method validation

5.2.1. Specificity

No significant interfering peak was observed at the retention times of the target insulin analogues in any of the six sources of biological matrices following PPT+MAX-SPE, confirming high method specificity.

5.2.2. Carry-over

No carry-over was detected in blanks injected after 10 and 50 ng/mL QCs. After 100 ng/mL QCs, trace signals (0.11 % – 0.13 % of the QC response) were observed in subsequent blanks, which were well below the LLOQ and considered analytically insignificant.

5.2.3. Matrix effect and extraction recovery

As summarized in [Table 2](#) (see also [Supplementary Materials Section S2](#) for aspart and glargine insulins analogues), the IS-NMF for human insulin ranged between the acceptable range of 85 % – 115 % across all matrices. In contrast, extraction recovery varied considerably among tissue types, ranging from 23.55 % to 60.16 %.

Extraction recovery is a critical factor governing method sensitivity

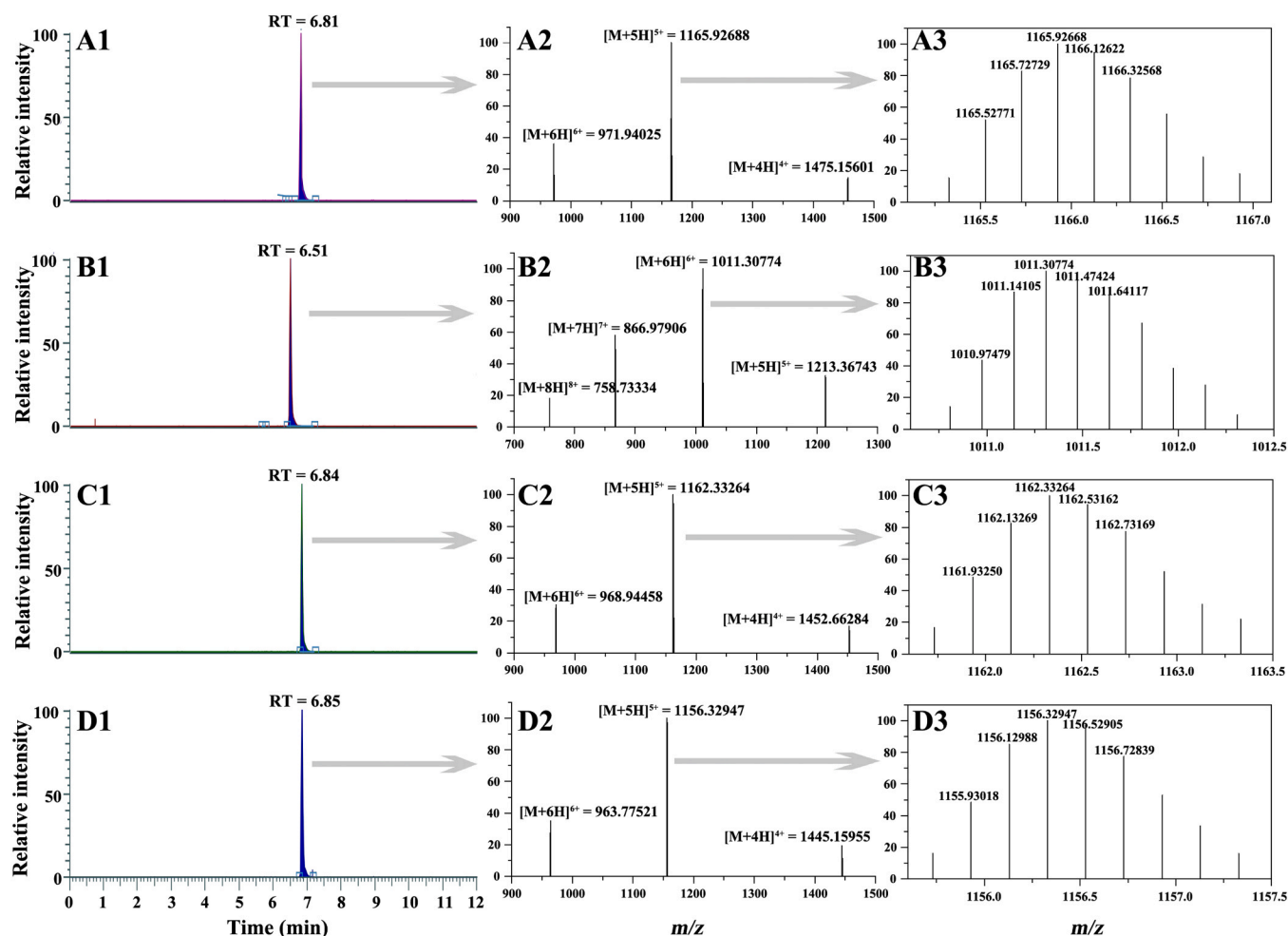


Fig. 1. Representative extracted ion chromatograms (EICs), charge state distributions, and isotope patterns of the four insulin analogues (each at 100 ng/mL in neat solution). (A, B, C, and D corresponded to insulin Aspart, Glargine, Human, and Porcine, respectively; Panels 1, 2, and 3 represented EICs, charge state distributions, and isotope distribution for each analogue).

and linearity; however, reported recovery rates for insulin vary substantially across methodologies. For instance, Liu *et al.* reported average recoveries of 91 % (IAP), 47 % (PPT + MCX-SPE), and 27 % (PPT + MAX-SPE) [23], while Bottinelli *et al.* observed 1.7 % – 58.8 % using ultrafiltration and IAP methods [24], consistent with the matrix-dependent variability observed in our study. Importantly, despite moderate and variable absolute recovery, the method achieved acceptable sensitivity and precision, confirming that recovery consistency sufficed reliable quantification. The physicochemical basis for these recovery limitations remains incompletely understood. Our results highlight two key influencing factors:

First, PPT solvent composition proved critical. While conventional protocols use $\geq 2:1$ (v/v) acidic organic solvent to sample [25], we found a 1:1 ratio of 80 % methanol (containing 0.1 % formic acid) to plasma significantly improved recovery (≥ 70 %). This may be attributed to insulin's polar residues reducing solubility in pure organic phases, alongside enhanced solubility near its isoelectric point ($pI = 5.3$) under acidic aqueous-organic conditions [26].

Second, competitive binding occurs during MAX-SPE. The sorbent combines reversed-phase and anion-exchange mechanisms [27], yet LC-HRMS analysis revealed numerous co-extracted endogenous peptides, particularly at low insulin concentrations, suggesting competition for anion-exchange sites, as supported by additional peaks in total ion chromatograms.

5.2.4. Linearity

The method demonstrated linear response for all target insulin analogues across the four postmortem matrices. A matrix-dependent LLOQ was observed: human insulin showed the widest linear range in plasma (0.1 – 100 ng/mL, LLOQ = 0.1 ng/mL), whereas liver, kidney, and adipose tissue exhibited linearity from 1 to 100 ng/mL (LLOQ = 1 ng/mL). Synthetic analogues (aspart and glargine) showed LLOQs consistent with human insulin in corresponding matrices. Calibration parameters, including equation, correlation coefficient, and linearity were compiled in [Supplementary Materials](#) Section S3.

5.2.5. Accuracy and precision

The method exhibited satisfactory accuracy and precision across all four matrices. For human insulin, both intra- and inter-accuracy ranged from 89.3 % to 106.9 %, with intra- and inter-day precision varied between 1.2 % and 11.1 %. All results met predefined validation criteria, confirming reliable quantitative performance.

5.3. Forensic application

The validated LC-Orbitrap-HRMS method was successfully applied to two independent forensic cases of suspected fatal insulin overdose. Case 1 involved a 22-year-old male who received an excessive abdominal injection (approx. 900 IU) of NovoRapid® (insulin Aspart) and NovoMix® 30 while case 2 involved a 60-year-old female with an administered overdose (approx. 300 IU) of NovoMix® 30 (detailed case

Table 1
Multiply charged precursor ions of the four kinds of insulin analogues.

Insulin analogues	Molecular weight (Da)	Charge state	Extracted <i>m/z</i> in full scan mode	Retention time (min)
Aspart	5826	[M+4H] ⁴⁺	1456.65698,	6.81
			1456.90515,	
			1457.15601 ,	
		[M+5H] ⁵⁺	1457.40723,	
			1457.65637	
			1165.52771,	
			1165.72729,	
		[M+6H] ⁶⁺	1165.92688* ,	
			1166.12622,	
			1166.32568	
Glargine	6062	[M+5H] ⁵⁺	971.60663,	6.51
			971.77307,	
			971.94025 ,	
			972.10675,	
		[M+6H] ⁶⁺	972.27319	
			1212.96997,	
			1213.16919,	
			1213.36743 ,	
		[M+7H] ⁷⁺	1213.56763,	
			1213.76855	
Human	5807	[M+4H] ⁴⁺	1010.97479,	6.84
			1011.14105,	
			1011.30774* ,	
			1011.47424,	
		[M+5H] ⁵⁺	1011.64117	
			866.69312,	
			866.83563,	
			866.97906 ,	
		[M+6H] ⁶⁺	867.12189,	
			867.26477	
Porcine	5777	[M+4H] ⁴⁺	758.48218,	6.85
			758.60773,	
			758.73334 ,	
			758.85730,	
		[M+5H] ⁵⁺	758.98413	
			1452.16406,	
			1452.41296,	
			1452.66284 ,	
		[M+6H] ⁶⁺	1452.91296,	
			1453.16394	

Notes: The bolded *m/z* represented the highest response abundance per charge. * represented the precursor ions applied in SIM mode.

information was provided in [Supplementary Materials](#) Section S3.)
As quantified in [Table 3](#), substantially elevated concentrations of insulin were detected in subcutaneous adipose tissue from the injection sites of both cases. Specifically, the method successfully identified and

quantified aspart and human insulin in adipose tissue, with human insulin also detected in the liver of case 1 (representative chromatograms were shown in [Supplementary Materials](#) Section S4). This distribution pattern matched the recovered insulin formulations and provided definitive evidence of exogenous administration, crucial for cause-of-death determination.

The significantly higher concentration in injection-site adipose tissue compared to liver and plasma aligns with the established local depot effect of subcutaneously injected insulin [28], confirming adipose tissue as the optimal forensic specimen for overdose confirmation. Our findings underscored the forensic necessity of analyzing adipose tissue from the suspected injection site, as insulin was uniquely detected at forensically conclusive levels in this matrix in Case 2, and at vastly higher concentrations in Case 1, whereas other matrices yielded negative or equivocal results. It was also noteworthy that in Case 1, the ratio of the rapid-acting insulin Aspart to Human insulin was much lower in the adipose tissue than the administered formulation, which may reflect the more rapid systemic dispersion and clearance of the rapid-acting analogue, a factor that could be critical for timing the injection event. However, it was important to acknowledge that the general applicability of our findings was based on the analysis of two fatal cases. While the results were compelling and demonstrated the method's practical utility, analysis of a larger cohort of suspected insulin intoxication cases was warranted to further establish the concentration ranges and distribution patterns in forensic practice.

While this method was developed and validated for postmortem matrices, its high sensitivity and specificity suggest potential utility in a clinical context for confirming suspected insulin overdose in living patients, provided that adipose tissue biopsies are obtainable. However, the invasive nature of tissue sampling makes plasma the primary clinical matrix; our method's LLOQ of 0.1 ng/mL for plasma is theoretically sufficient to detect supra-therapeutic concentrations, though its clinical application would require further validation with antemortem samples. Beyond overdose confirmation, the method's core advantage of distinguishing exogenous insulin analogues from endogenous human insulin opens significant avenues for precise pharmacokinetic studies. For instance, it can accurately characterize the absorption and distribution profiles of insulin analogues without interference from the endogenous hormone.

The comparative analysis with the CLIA revealed the complementary strengths and limitations of each technique. While the CLIA method demonstrated high sensitivity by detecting immunoreactive insulin in all specimens [29], it lacked the specificity to differentiate between insulin analogues. In contrast, the superior specificity of LC-Orbitrap-HRMS was essential for definitive analogue identification. However, a notable discrepancy was observed: LC-Orbitrap-HRMS returned negative results in Case 1 plasma and Case 2 liver/plasma, specimens that were positive by CLIA. This discrepancy was likely multifactorial, attributable to the insulin concentrations in those specimens falling below the matrix-specific LLOQs of the LC-Orbitrap-HRMS method, a phenomenon potentially influenced by postmortem redistribution, insulin degradation in vascular compartments, and others [30]. The positive CLIA results in these samples may also be partly explained by the assay's known cross-reactivity with insulin metabolites [31]. Besides, a primary limitation of this study was the restricted understanding of comprehensive insulin distribution due to the limited tissue sampling (adipose tissues, liver, and plasma). To fully elucidate postmortem insulin pharmacokinetics, future investigations should prioritize the systematic collection of a wider range of tissues, such as brain, kidney, and muscle.

Consequently, we proposed an integrated analytical strategy: initial screening using CLIA followed by confirmatory LC-HRMS analysis, particularly for adipose tissue quantification and analogues differentiation. This approach combines CLIA's sensitivity with LC-HRMS's specificity, supplemented by injection-site histopathology and C-peptide/insulin ratio analysis for comprehensive death investigation [32].

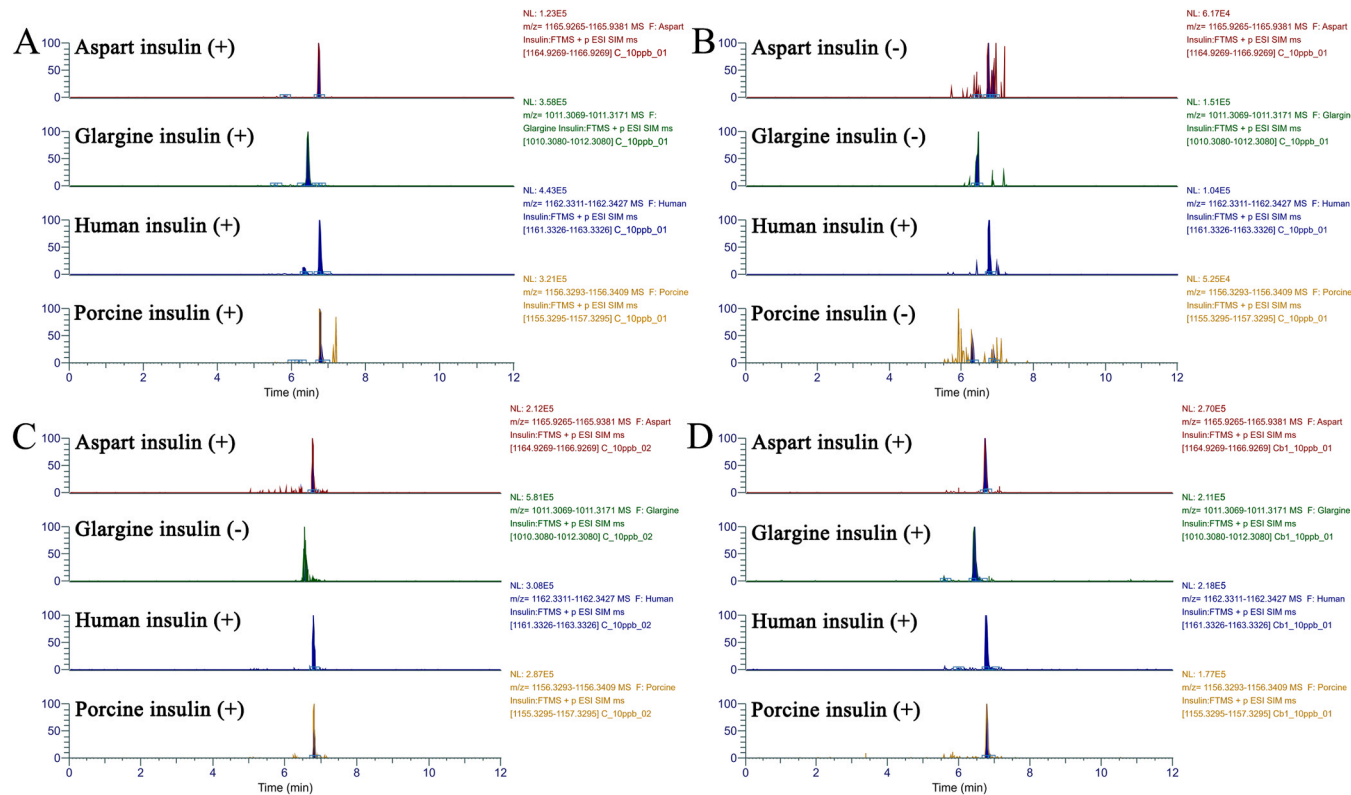


Fig. 2. Comparison of sample preparation methods for extracting insulin (spiked concentration: 10 ng/mL) from various biological matrices. Representative EICs were shown for: (A) plasma processed by PPT; (B) liver homogenate processed by PPT; (C) liver homogenate processed by PPT+MCX-SPE; and (D) liver homogenate processed by PPT+MAX-SPE (six independent replicates per matrix and method). The symbols “+ ” and “–” indicate the presence or absence of a detectable insulin peak, respectively.

Table 2
Method validation results for human insulin in four postmortem matrices.

Matrix	Concentrations (ng/mL)	IS-NMF Mean ± RSD (%)	ER Mean ± RSD (%)	Intra-day (n = 6)		Inter-day (n = 18)	
				Accuracy (%)	Precision (RSD %)	Accuracy (%)	Precision (RSD %)
Plasma	0.1	94.5 ± 8.3	23.1 ± 2.7	90.1	2.5	93.8	2.9
	1	97.9 ± 6.4	27.6 ± 2.6	95.4	4.5	97.8	1.4
	10	98.8 ± 2.3	28.6 ± 2.9	106.9	1.9	104.5	3.9
	100	95.1 ± 1.1	29.5 ± 3.6	99.1	1.2	100.2	5.1
Liver	1	99.7 ± 3.6	47.4 ± 13.2	89.3	6.5	89.5	11.1
	10	90.6 ± 3.4	50.1 ± 6.7	102.2	1.8	103.4	3.3
	100	99.3 ± 3.6	60.2 ± 5.2	98.5	2.2	101.9	5.3
Kidney	1	99.0 ± 3.0	24.2 ± 7.1	95.2	5.7	96.3	7.3
	10	98.1 ± 2.4	24.6 ± 5.4	99.2	8.7	100.1	9.3
	100	98.7 ± 3.4	23.6 ± 2.8	103.9	6.7	99.1	5.4
Adipose tissue	1	97.1 ± 3.1	26.8 ± 2.1	96.7	2.9	94.1	2.1
	10	94.2 ± 1.7	26.4 ± 5.5	98.1	2.3	98.3	4.6
	100	95.7 ± 4.4	27.8 ± 5.0	100.2	1.8	99.6	5.1

Table 3
Insulin concentrations detected by CLIA and LC-Orbitrap-HRMS methods.

Case	Matrix	Concentration of insulins		
		CLIA (μIU/mL)	LC-Orbitrap-HRMS (ng/mL)	
			Aspart	Human
1	Adipose tissue	25470.0	5.3	743.1
	Liver	150.1	\	4.8
	Plasma	15.6	\	\
2	Adipose tissue	1781.0	53.2	1.1
	Liver	100.9	\	\
	Plasma	0.6	\	\

Notes: “\” indicates a negative test result.

6. Conclusion

The presented study focused on the development and validation of an LC-Orbitrap-HRMS method for the quantification of human insulin and selected synthetic analogues in plasma, liver, kidney, and adipose tissue. By optimizing a combined PPT and MAX-SPE sample preparation approach, the method enabled simultaneous insulin quantification across these diverse tissue types. Method validation results demonstrated acceptable linearity, accuracy, precision, and sensitivity suitable for forensic applications. When applied to two insulin-related fatalities, high concentrations of Aspart and Human insulins were successfully detected and quantified in subcutaneous adipose tissue from injection sites. These findings provided definitive evidence of exogenous insulin administration, supporting accurate cause-of-death determination in

forensic investigations.

CRediT authorship contribution statement

Gao Xiangting: Writing – review & editing, Supervision, Resources, Conceptualization. **Gao Hang:** Writing – review & editing, Validation, Methodology. **Tao Jing:** Writing – review & editing, Resources. **Zhang Jiaxin:** Writing – review & editing, Supervision, Methodology. **Yuan Yuhao:** Writing – review & editing, Validation, Resources, Investigation. **Yu Zhonghao:** Writing – original draft, Validation, Methodology, Investigation, Data curation. **Zhou Yiwu:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Conceptualization.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jpba.2025.117282](https://doi.org/10.1016/j.jpba.2025.117282).

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